

- 2 Pepsin is an enzyme that hydrolyses proteins (protease). Some students used pepsin from the stomach of a mammal.

The activity of the pepsin was investigated by placing a small quantity of the enzyme with a known concentration of the protein albumen.

Fig. 2.1 shows the progress of the enzyme-catalysed reaction that was carried out at 20 °C.

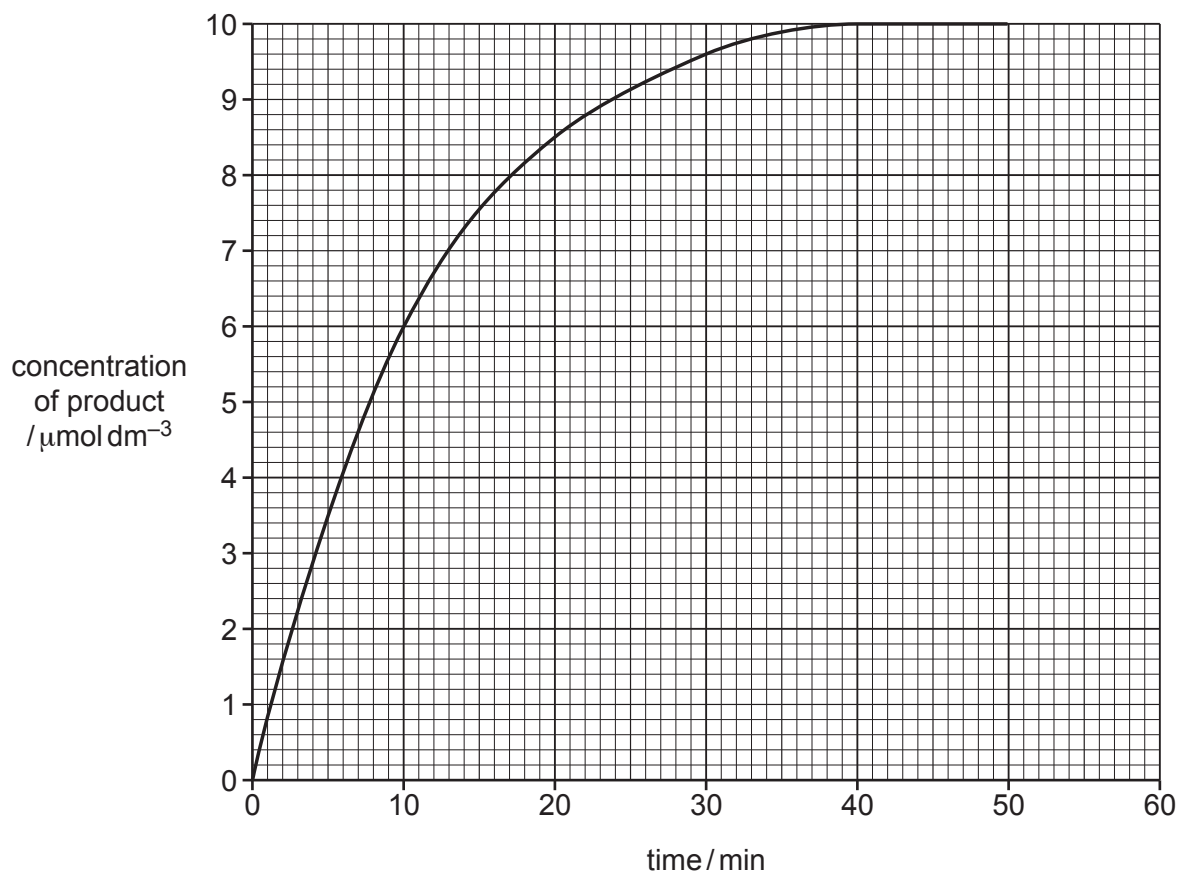


Fig. 2.1

- (a) Calculate the initial rate of the reaction.

initial rate of reaction = [2]

- (b) (i) The procedure was repeated to find the effects on the activity of the pepsin using a competitive inhibitor at the same temperature, 20 °C.

Predict the results that will be obtained using the competitive inhibitor.

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- (ii) The procedure was repeated **without** the competitive inhibitor but at the higher temperature of 30 °C.

Predict the results that will be obtained at 30 °C.

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- (c) The students extended their investigation by using pepsin from a different species of mammal. The experiments were carried out at 20 °C and **without** a competitive inhibitor.

With reference to Fig. 2.1, explain the advantage of calculating the initial rate of reaction in each experiment.

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[Total: 8]